Week 08: Normalization

Learning outcomes: CLO2, CLO3

Lesson

- **Functional dependency** X → Y: one X gives one Y. Here `province_pcode → province_name, area_km2`.
- The example's flat report repeats province data on every commune row. As the only copy it would cause anomalies:
- **update:** change a name in many rows;
- **insert:** can't add a province with no communes;
- **delete:** deleting the last commune loses the province.
- **1NF:** one value per cell.
- **2NF:** 1NF + non-key columns depend on the whole key. E.g. key (`student_id`, `course_code`) with `course_name` breaks 2NF. Our tables have one-column keys.
- **3NF:** 2NF + no non-key → non-key dependency. `commune_pcode → district_pcode → province_name` is transitive, so split the report.
- **BCNF:** for every X → Y (Y not in X), X is a superkey.
- A flat (denormalized) report is fine for reading if you can rebuild it.

Worked example

Run `example.sql` (how (`setup/README.md`)), then answer in `work.sql`.

Tables for this week

Read `cambodia`, write `student_work`. All columns are `NOT NULL`. Samples are the first two rows by code.

`cambodia.province`

Column	Type	Key	Meaning
`province_pcode`	`CHAR(4)`	PK	Province code
`province_name`	`VARCHAR(100)`		Province name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place

Sample rows:

province_pcode	province_name	area_km2
KH01	Banteay Meanchey	6145.9
KH02	Battambang	11869.0

`cambodia.district`

Column	Type	Key	Meaning
`district_pcode`	`CHAR(6)`	PK	District code
`district_name`	`VARCHAR(100)`		District name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`province_pcode`	`CHAR(4)`	FK	Province code

Sample rows:

district_pcode	district_name	area_km2	province_pcode
KH0102	Mongkol Borei	807.2	KH01
KH0103	Phnum Srok	761.7	KH01

cambodia.commune

Column	Type	Key	Meaning
commune_pcode	CHAR(8)	PK	Commune code
commune_name	VARCHAR(100)		Commune name (English)
area_km2	DECIMAL(12,1)		Area in km², one decimal place
district_pcode	CHAR(6)	FK	District code

Sample rows:

commune_pcode	commune_name	area_km2	district_pcode
KH010201	Banteay Neang	52.0	KH0102
KH010202	Bat Trang	40.3	KH0102

Join keys: `district.province_pcode = province.province_pcode`; `commune.district_pcode = district.district_pcode`.

Report columns: `commune_pcode`, `commune_name`, `commune_area_km2`, `district_pcode`, `district_name`, `district_area_km2`, `province_pcode`, `province_name`, `province_area_km2`. One row per commune; a query result, not a table.

More: data guide (`data/README.md`).

Lab tasks

1. Run the report in `example.sql`; save its columns and output.
2. Write the functional dependencies and a candidate key.
3. Use repeated province values to explain update, insert and delete anomalies. Don't change the source.
4. Split the report into province, district and commune tables with keys.
5. Join them back and compare with the report: same row count, same values.

Homework

1. Explain 1NF, 2NF, 3NF and BCNF with this data.
2. When is a flat report useful? How would you refresh it?

Capstone sprint

C8 / Checkpoint 2: Submit schema v1: ERD, dependencies, migration SQL, import script. Justify any denormalization.

Submit

Answers in `work.sql`, reflection in `reflection.md`, output in `results/`. Commit and push. See SUBMISSIONS.md.